

A systematic analysis of representations in artificial neural networks across hyperparameters

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Abstract

Artificial neural networks (ANNs) are indispensable tools for modeling human cognition, yet their internal representational geometry remains poorly understood. Researchers often select model hyperparameters based on task performance, but this practice assumes that high accuracy implies biological relevance. However, if networks with comparable performance develop different internal structures, accuracy-based selection may lead to misleading comparisons with brain data. Building on prior studies, which have usually focused on single hyperparameters, we provide a systematic multi-factor analysis of representational variability using a fully crossed hyperparameter design. This study explored the influence of eight hyperparameters – batch size, hidden layer size, learning rate, nonlinearity, optimizer, L1 and L2 regularization, and weight initialization – on the representations learned by 1504 feedforward networks trained on a dual-task MNIST dataset. Using representational similarity analysis, we quantified the consistency of the representations of the networks. Spearman rank correlations between representational dissimilarity matrices revealed substantial representational variability ($\rho = .55$), which decreased when averaging across networks with identical hyperparameter configurations ($\rho = .70$), indicating that representational differences arose from both hyperparameter choices and training stochasticity (random initialization and batch sampling). In addition, we ran a principal component analysis, which suggests that activation function, hidden layer size, optimizer choice and learning rate had the largest influence on network representations. Together, these findings show that networks with comparable behavioral performance can develop substantially different internal representations as a function of hyperparameter choice. More broadly, the results demonstrate that hyperparameters matter for representational geometry in complex and interacting ways that are not fully captured by task performance alone. This suggests that implementation details can impact inferences on model-brain comparisons, as representational conclusions may vary across different hyperparameter settings.

Introduction

Artificial neural networks (ANNs) are increasingly used as computational models to study neural data, motivated by the idea that artificial and biological systems may rely on similar principles of information processing. Both are thought to encode information in distributed patterns of activity across populations of units rather than localized

representations (Yamins & DiCarlo, 2016). Representational similarity analysis (RSA) enables direct comparison between representations derived from ANNs and neural data obtained with techniques such as fMRI and EEG (Kriegeskorte & Kievit, 2013; Kriegeskorte et al., 2008). By focusing on the geometry of internal activation patterns, RSA enables comparisons between model representations, brain regions and cognitive variables beyond performance-level metrics (e.g., Cichy et al., 2016; Colin et al., 2025; Holroyd et al., 2018; Khaligh-Razavi et al., 2017).

Despite the growing use of RSA with neural networks, relatively little is known about how training hyperparameters influence the representational geometry that emerges inside these models. Hyperparameters are often tuned to maximize generalization performance (Goodfellow et al., 2016), yet networks with comparable task accuracy can still develop substantially different internal representations (Mehrer et al., 2020). This poses a challenge for RSA-based model-brain comparisons, which depend directly on representational structure. Sensitivity to hyperparameters therefore raises concerns about representational stability, interpretability and replicability.

Most prior work on hyperparameters has focused on performance outcomes (e.g., Cooney et al., 2020; Liao et al., 2022; Pinto et al., 2009; Roy et al., 2023). Studies addressing representational geometry typically manipulate a single hyperparameter while holding others constant, such as weight initialization (Flesch et al., 2022; Li et al., 2016; Mehrer et al., 2020; Wang et al., 2018), hidden layer size and depth (Nguyen et al., 2020), nonlinearity (Alleman et al., 2024) and optimization algorithms (Chaudhury & Yamasaki, 2021; Zhou et al., 2020). Although these studies demonstrate that individual hyperparameters can affect representations, results vary substantially across architectures and tasks, highlighting the limitations of single-factor approaches and the potential importance of interactions between hyperparameters.

In this study, we examine how multiple hyperparameters jointly shape representational geometry in a feedforward neural network trained on the MNIST dataset. The network performed two binary digit classification tasks (even/odd and small/large) to promote task-relevant representations. Using a fully crossed hyperparameter grid, we varied batch size, hidden layer size, weight initializa-

tion, learning rate, nonlinearity, optimizer type, L1 and L2 regularization. Representational dissimilarity matrices derived from network activations were compared using principal component analysis (PCA). This systematic approach provides insight into the stability and reliability of network representations and their suitability for RSA-based neuroscience applications. To our knowledge, the approach used in the present work—a systematic study of how hyperparameter variations affect representations for a fixed task—is novel in the machine learning literature.

Methods

Network Architecture and Task Design

Feedforward neural networks were trained on a modified version of the MNIST handwritten digit dataset (28×28 grayscale images, digits 0–9; Deng, 2012; LeCun et al., 1998). Each stimulus was evaluated under two binary task contexts: small/large (digits < 5 vs. ≥ 5) and even/odd. This dual-task design ensured that the problem was not linearly solvable and encouraged the emergence of more interpretable internal representations. Task context was provided via an additional binary input unit, yielding 785 input units in total. The networks contained a single hidden layer and two output units. For both task contexts, the first output unit encoded “small” or “even” and the second “large” or “odd”, corresponding to target vectors [1,0] and [0,1], respectively. Networks were trained using early stopping: training of an individual network terminated when validation loss failed to improve by at least 0.01 for three consecutive epochs, with a maximum of 100 epochs. Networks that failed to reach a minimum classification accuracy of 90% were excluded from further analysis.

Hyperparameter Grid Search and Model Selection

A fully crossed hyperparameter grid search was conducted across batch size, hidden layer size, weight initialization, learning rate, activation function, optimizer type, L1 and L2 regularization. Hyperparameter values were selected to span qualitatively distinct training regimes, yielding 576 configurations (Table 1). Each configuration was trained eight times, for a planned total of 4608 networks. To improve computational efficiency, if two consecutive networks in a configuration both failed to reach the 90% accuracy criterion, the entire configuration was rejected and no further networks were trained for that configuration.

With respect to weight initialization, we manipulated the initialization scale by multiplying the standard Xavier or He initialization by a fixed factor. Xavier initialization (Glorot & Bengio, 2010) was used for tanh and sigmoid, and He initialization (He et al., 2015) for ReLU networks, with scale factors of 1.0 or 0.01. This manipula-

Table 1: Hyperparameter values used in the grid search.

Hyperparameter	Values
Batch size	1, 32
Hidden layer size	16, 128
Weight initialization factor	1.0, 0.01
Learning rate	0.00001, 0.001, 0.1
Nonlinearity	ReLU, tanh, sigmoid
Optimizer	Adam, SGD
L1 regularization	0.0, 0.0001
L2 regularization	0.0, 0.0001

tion was motivated by theoretical work distinguishing rich and lazy learning regimes, in which smaller initialization scales constrain feature learning relative to larger scales (Chizat et al., 2018).

Representational Dissimilarity Matrices

Representational structure was analyzed at the level of hidden-layer activations. For each network, activation patterns were extracted for a balanced test set of 100 stimuli (5 exemplars × 10 digits × 2 task contexts). Each stimulus was presented under both task contexts (small/large and even/odd), resulting in representations for every digit–task combination. Pairwise cosine distances between activation vectors were computed to form RDMs (100 × 100). These cosine-distance RDMs formed the basis for the analyses. A mean RDM across all raw RDMs is shown in Figure 1. To compare representational geometry across networks, each RDM was vectorized by extracting the upper-triangular entries (excluding the diagonal), producing a feature vector of stimulus-pair dissimilarities per network.

Representational Similarity Across Networks

Representational similarity was assessed via pairwise Spearman correlations between vectorized RDMs, calculated both for individual networks and for configuration-averaged RDMs to dissociate stochastic from hyperparameter-driven variability.

Principal Component Analysis of RDM Structure

To identify dominant axes of variation in representational geometry, PCA was applied to the matrix of vectorized RDMs (networks × stimulus-pair features). Prior to PCA, each RDM vector was row-wise z-scored to remove differences in overall dissimilarity magnitude while preserving relative distance structure, a normalization necessary given differences in activation ranges across nonlinearities (ReLU vs. sigmoid/tanh). Analyses focused on the leading components explaining the largest fractions of variance. To relate principal components to hyperparameters, one-way ANOVAs were performed on

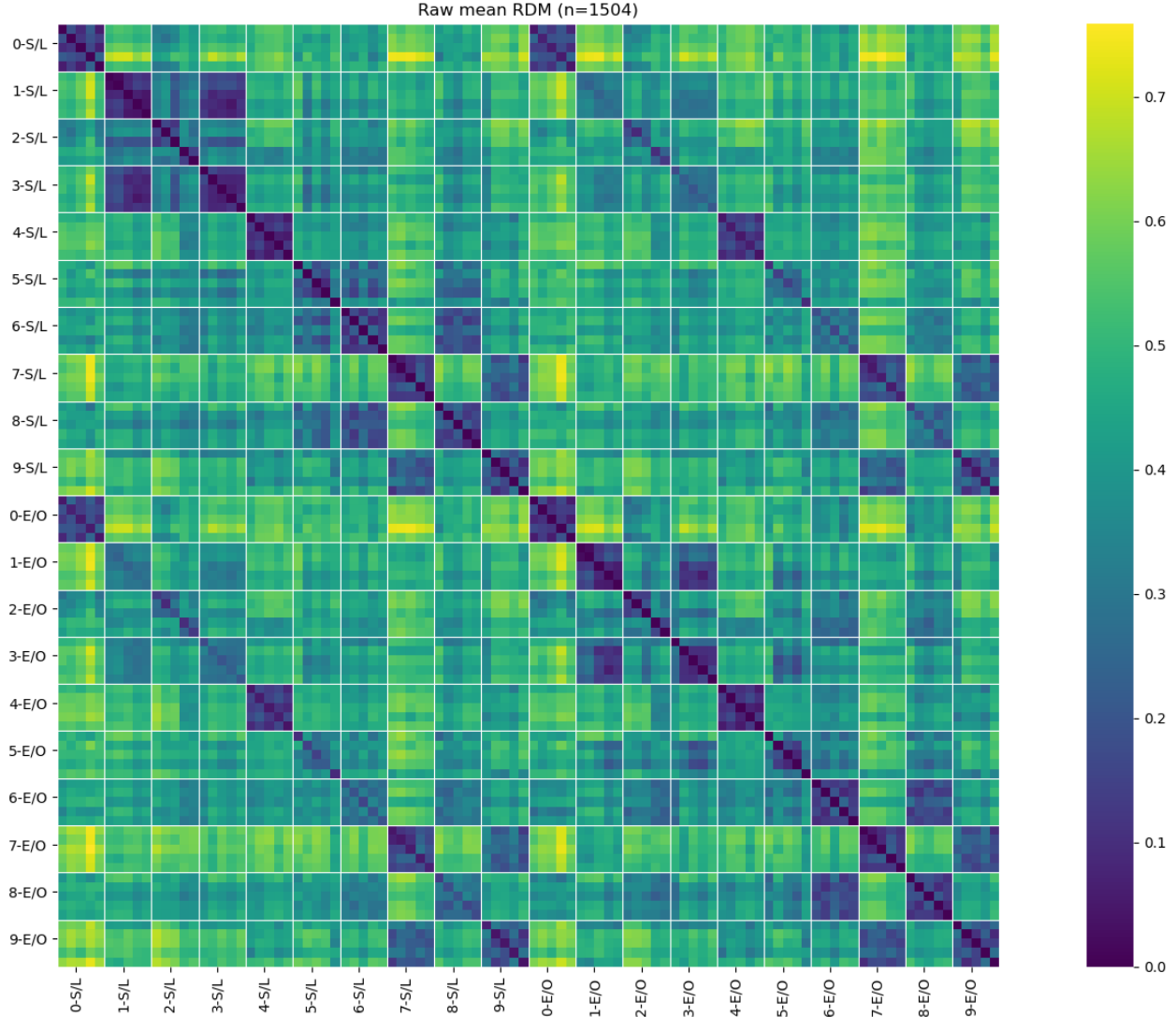


Figure 1: Mean cosine RDM (100×100) across all networks ($n = 1504$) computed by element-wise averaging. Stimuli are ordered by task (Small/Large; Even/Odd) and digit, with each labeled block representing five different exemplars of a digit within a task context (e.g., “0–S/L”). The diagonal indicates zero self-distance. Lighter colors indicate greater representational dissimilarity (cosine distance closer to 1), while darker colors indicate greater similarity (cosine distance closer to 0).

Results

Training Success Rates

Of the 4608 planned training runs, 1504 networks (32.6%) met the 90% accuracy criterion and were retained for analysis. This high failure rate is expected in a fully crossed hyperparameter grid, as many configurations fall outside stable training regimes (Bergstra & Bengio, 2012). Table 2 identifies hyperparameter combinations that were incompatible with successful training: none of the 1504 successful networks contained any of these combinations. Any configuration includ-

component scores for each hyperparameter, with effect sizes quantified using η^2 , and robustness assessed using permutation tests (1000 label shuffles). To aid interpretation in stimulus space, an RDM-based visualization was derived in which, for each of PC1–PC3, a PC-extremes difference RDM was computed by subtracting the mean RDM of networks in the bottom 20% of scores from that of the top 20%. Finally, to summarize recurring representational regimes, k-means clustering was applied to PCA scores, with the number of clusters determined using the elbow criterion.

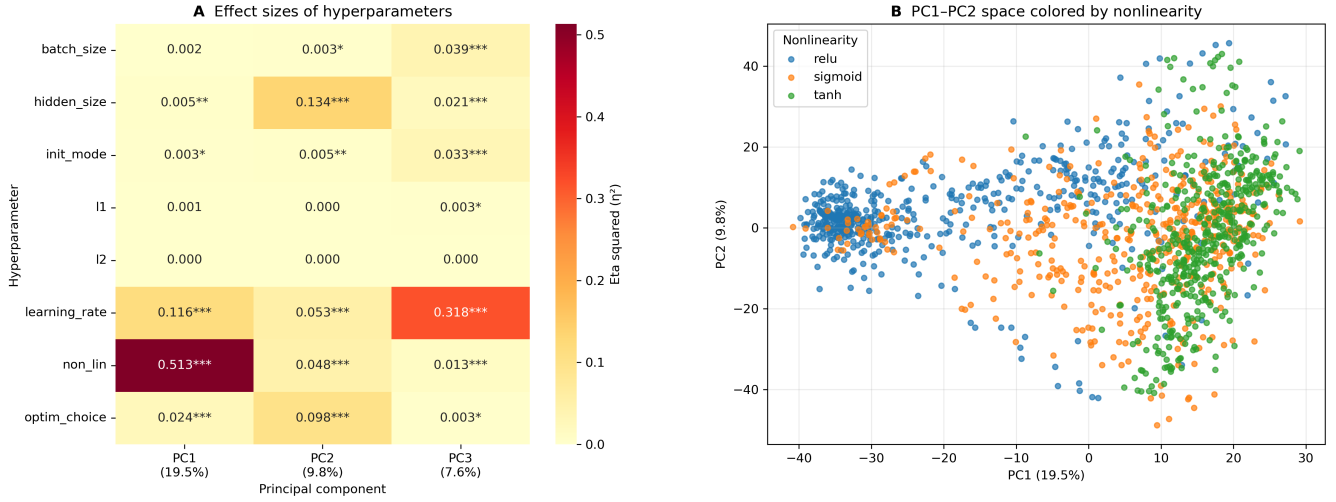


Figure 2: Hyperparameter contributions to principal components. (A) Effect sizes (η^2) from one-way ANOVA analyses, quantifying the fraction of variance in principal component scores explained by each hyperparameter. η^2 values reflect the relative contribution of individual hyperparameters to the structure captured by each component. (B) Projection of individual networks onto PC1–PC2 space, with networks colored by nonlinearity.

Table 2: Failed hyperparameter combinations

Batch size 1 + learning rate 0.1
Learning rate 0.1 + tanh
Learning rate 0.00001 + SGD
Learning rate 0.1 + Adam

ing these combinations failed to meet the performance criterion, regardless of other hyperparameter settings.

Similarity Among the RDMs

Individual RDMs Pairwise comparisons among the 1504 individual RDMs revealed moderate similarity in representational geometry (ρ range = -.27 to .99; mean ρ = .55, SD = 0.16). A small proportion of comparisons (3%) yielded negative correlations, indicating inversely structured representations for a minority of network pairs.

Configuration-Averaged RDMs Averaging RDMs across repeated runs within each hyperparameter configuration increased similarity (ρ range = .21 to .99; mean ρ = .70, SD = 0.13), with no negative correlations observed.

Principal Component Analysis

PCA of the 1504 individual RDMs revealed three principal components explaining 36.9% of the total variance (PC1: 19.5%, PC2: 9.8%, PC3: 7.6%). ANOVAs identified the main hyperparameter drivers of each component: PC1 was dominated by activation function (η^2 = .51) and learning rate (η^2 = .12), PC2 by hidden layer

size (η^2 = .13) and optimizer (η^2 = .10), with smaller contributions from learning rate and activation function (both η^2 = .05); and PC3 primarily by learning rate (η^2 = .32) (Fig. 2A).

PC1: Nonlinearity PC1 (19.5% explained variance) was primarily driven by nonlinearity (η^2 = .51). In the PC1–PC2 projection (Fig. 2B), networks formed a continuous gradient, with ReLU networks at low PC1 values, tanh networks at high values, and sigmoid networks occupying an intermediate region with partial overlap, indicating graded rather than discrete separation. To characterize representational differences along PC1, networks in the lowest and highest 20% of PC1 scores were contrasted using a difference RDM (Fig. 3). This RDM was uniformly negative, indicating higher overall dissimilarity for ReLU networks relative to tanh and sigmoid networks. However, a moderate rank-order correlation between the two groups (ρ = .57) indicates that nonlinearity affected not only dissimilarity magnitude but also the relative configuration of stimulus relationships. Together, these findings identify nonlinearity as a major determinant of representational geometry, influencing both overall dissimilarity and the organization of stimulus representations.

PC2: Hidden Layer Size and Optimizer PC2 (9.8% explained variance) was primarily associated with hidden-layer size (η^2 = .13) and optimizer choice (η^2 = .10). Variation along PC2 was driven mainly by tanh and sigmoid networks, as ReLU networks clustered near zero on this axis and were largely absent from the top and bottom 20% extremes (Fig. 2B). In contrast to

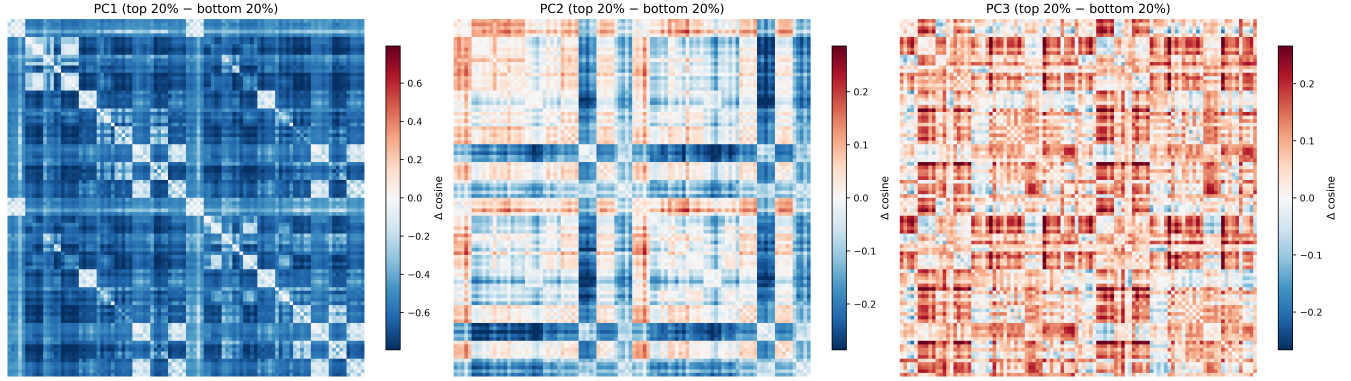


Figure 3: Representational variation along principal components. Difference RDMs contrasting networks in the top and bottom 20% of scores along PC1 (19.5%), PC2 (9.8%) and PC3 (7.6%) (high–low). Rows and columns follow the stimulus ordering described in Fig. 2. Color values reflect differences in cosine dissimilarity.

PC1, the PC2 difference RDM showed both positive and negative values, indicating stimulus-specific rather than global representational changes (Fig. 3).

These effects were strongest for digits 7 and 9, which showed markedly higher dissimilarity in networks with low PC2 scores, and for digit 0, which exhibited the opposite pattern (Fig. 3). Despite these localized differences, overall rank-order structure remained moderately preserved ($\rho = .74$), indicating reorganization of specific stimulus relationships rather than wholesale restructuring. Inspection of hyperparameter composition revealed that networks in the bottom 20% of PC2 scores were dominated by small networks (hidden size 16: 85%) trained with Adam (83%), whereas networks in the top 20% had a higher proportion of large networks (hidden size 128: 69%) trained with SGD (62%). Together, these results indicate that hidden-layer size and optimizer primarily shape representational geometry through stimulus-specific contrasts rather than uniform scaling.

PC3: Learning Rate PC3 (7.6% explained variance) was primarily associated with learning rate ($\eta^2 = .32$). The PC3 difference RDM showed predominantly positive values, indicating higher overall dissimilarity across nearly all stimulus pairs for networks with high PC3 scores (Fig. 3). Networks in the bottom 20% of PC3 scores were predominantly trained with a moderate learning rate (0.001: 79%), whereas the top 20% showed a higher proportion of networks trained with a lower learning rate (0.00001: 63%). Despite these magnitude differences, rank-order structure was strongly preserved ($\rho = .78$), indicating that learning rate mainly modulates the scale of representational distances rather than their relative configuration, making PC3 the most scale-like component.

K-Means Clustering in Principal Component Space

To provide a discrete summary of the representational structure revealed by PCA, networks were clustered in the three-dimensional space defined by PC1–PC3 using k-means clustering. This yielded four clusters of comparable size: Cluster 1 ($n = 402$, 26.7%), Cluster 2 ($n = 324$, 21.5%), Cluster 3 ($n = 388$, 25.8%), and Cluster 4 ($n = 390$, 26.0%) (Fig. 4). Cluster separation was aligned with the principal component directions. Thus, clustering discretizes the continuous PCA gradients into a small number of representative groups without introducing additional dimensions of representational variation.

Discussion

The present study examined how hyperparameters shape the representational geometry of artificial neural networks. Using a fully crossed hyperparameter grid search in feedforward networks trained on a dual-task, we show that network representations are broadly similar yet exhibit systematic, structured variation driven by specific hyperparameter choices. Importantly, these representational differences arise among networks that are all capable of solving the tasks successfully, achieving a minimum test accuracy of 90%, demonstrating that comparable accuracy can mask substantial differences in internal representational organization.

Task Performance Permits Diverse Representational Geometries

Across individual networks, representational geometry exhibited moderate overall similarity, with a mean Spearman rank correlation of $\rho = .546$. This indicates the presence of a shared representational core across networks trained on the same task, reflecting common

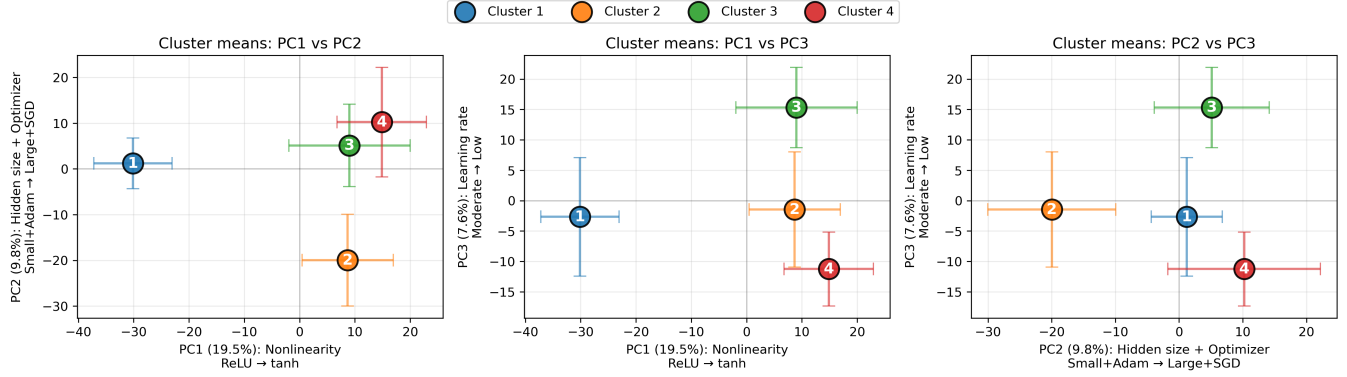


Figure 4: Cluster centroids projected onto pairs of principal components (PC1–PC3). Error bars indicate ± 1 SD. Axis labels indicate the dominant hyperparameter gradients associated with each principal component.

task-related constraints on representational organization. Such partial convergence is consistent with prior work showing that task optimization constrains representational geometry without uniquely determining it, leading networks trained on identical tasks to develop broadly similar yet non-identical representations (Colin et al., 2025; Mehrer et al., 2020). At the same time, similarity values spanned a wide range, from moderately negative to near-perfect correlations, revealing substantial variability in how individual networks organized stimulus relationships. The occurrence of negative correlations, although rare (3%), indicates that some networks converged on qualitatively different, and even inversely related, representational organizations, despite all networks being capable of solving the task at high accuracy. Together, these findings support the notion that multiple distinct representational geometries can underlie successful task performance.

Averaging RDMs across repeated training runs within each hyperparameter configuration increased overall similarity and eliminated negative correlations ($\rho = .70$). This pattern suggests that a considerable portion of representational divergence arises from stochastic factors inherent to training, such as random initialization and minibatch sampling, rather than from systematic hyperparameter effects per se. This interpretation is consistent with prior work showing that stochastic initialization alone can give rise to substantial differences in internal representations, even when network architecture and task performance are held constant (Li et al., 2016; Wang et al., 2018). In particular, RSA-based analyses have demonstrated that such initialization-driven variability leads to pronounced individual differences in representational geometry across network instances (Mehrer et al., 2020). However, stochasticity in learning is itself shaped by hyperparameter choice: parameters such as batch size, learning rate and optimizer modulate the magnitude and structure of gradient noise and thereby condition the variability of representational out-

comes across runs (Mandt et al., 2017). Consequently, part of the observed between-network variability reflects hyperparameter-dependent stochasticity. Nevertheless, meaningful differences persisted among the averaged RDMs, indicating that hyperparameters exert a reliable influence on representational geometry beyond stochastic variation alone.

Principal Components Reveal Structured, Task-Independent Representational Variation

The first three principal components together accounted for 36.9% of the variance in representational geometry, indicating the presence of dominant axes of variation alongside considerable residual variability. These components captured distinct forms of variation, ranging from global shifts in dissimilarity magnitude to localized, stimulus-specific reweighting. Importantly, this variation generalized across task contexts: neither the raw mean RDM nor the PC RDMs showed systematic differences between tasks (Figs. 1 and 3), as such effects would be visible as global shifts between within-task and cross-task quadrants of the matrix. This suggests that the principal components primarily reflect stimulus-level rather than task-specific structure.

The first principal component was strongly associated with nonlinearity ($\eta^2 = 0.51$), separating ReLU from tanh networks along a continuous axis. ReLU networks exhibited uniformly higher dissimilarity across all stimulus pairs, yet the moderate rank-order correlation between PC1 extremes (Spearman $\rho = 0.57$) indicates that this effect extends beyond simple uniform scaling. This pattern likely reflects ReLU’s unbounded activation and hard thresholding at zero, which generate greater variance and sparsity compared to the saturating ranges of tanh and sigmoid (Glorot & Bengio, 2010; Glorot et al., 2011). While ReLU amplifies representational distances, it does not uniformly scale all stimulus relation-

ships. Some pairs are disproportionately affected, leading to partial reordering of representational space. This highlights that nonlinearity influences both the overall magnitude and the structure of internal representations.

The second principal component captured stimulus-specific representational variation, driven by hidden layer size and optimizer type ($\eta^2 = .13$ and $.10$, respectively). This variation was restricted to tanh and sigmoid networks, as ReLU networks clustered near zero on PC2 (Fig. 2B). Despite localized differences, the rank-order correlation between PC2 extremes (Spearman $\rho = .74$) indicates that overall representational structure remained largely preserved. Variation along PC2 was dominated by specific stimulus relationships, most visibly involving digits 7 and 9. Notably, digits 7 and 9 are functionally equivalent (both “large” and “odd”), yet small networks trained with Adam exhibited higher dissimilarity between these digits compared to large networks trained with SGD. This suggests that the observed separation is not task-driven but reflects interactions between capacity and optimization within tanh and sigmoid networks. One possibility is that smaller networks rely more heavily on a restricted set of discriminative features that suffice for task performance but exaggerate distinctions between visually distinct stimuli, although this interpretation remains speculative. Together, these findings illustrate how hyperparameter choices can induce structured yet task-independent representational variation.

The third principal component was driven primarily by learning rate ($\eta^2 = .32$). As with PC2, variation along PC3 was largely absent in ReLU networks, which clustered near zero on this axis (Fig. 2B). PC3 differentiated clusters 3 and 4, both of which score high on PC2 and primarily comprise large networks trained with SGD (Fig. 4). Importantly, RDMs at opposing extremes of PC3 showed a strong rank-order correlation (Spearman $\rho = .78$), indicating that variation along this axis preserves the relative structure of representational distances while modulating their overall magnitude. This pattern suggests that PC3 captures predominantly scale-like variation, rather than changes in representational geometry. Consistent with this interpretation, prior work has shown that learning rate primarily affects the dispersion of representations without substantially altering their relational organization (Mandt et al., 2017).

Limited Contribution of Weight Initialization

In the present analysis, variation in weight initialization scale did not account for a substantial fraction of variance in representational geometry. Although discussions of rich–lazy learning often emphasize initialization scale—where large initial weights are associated

with near-linear, “lazy” behavior and smaller weights with “richer” feature learning (Chizat et al., 2018; Jacot et al., 2018)—recent work suggests that the rich–lazy distinction emerges from interactions among multiple aspects of the training regime, including learning rate, architecture, loss scaling, and optimization dynamics (Chizat et al., 2018; Chou et al., 2025). Moreover, the initialization scales explored here did not span the extreme regimes considered in theoretical treatments approaching the lazy limit, which often involve scale factors several orders of magnitude larger (e.g., 10^3 – 10^6 ; Chizat et al., 2018). From this perspective, the limited role of weight initialization observed here is not unexpected, and the rich–lazy learning distinction may still provide a meaningful framework for interpreting some representational differences across networks. However, the present results indicate that initialization scale alone does not drive large-scale variation in final representational geometry. Because the current analysis focuses on end-state representations rather than learning dynamics—where rich and lazy regimes are most clearly distinguished—initialization-dependent effects may be attenuated at convergence (Chou et al., 2025; Saxe et al., 2019).

Minimal Impact of L1 and L2 Regularization

L1 regularization penalizes absolute weight magnitude and promotes sparsity, whereas L2 penalizes squared weight magnitude and discourages large weights (Goodfellow et al., 2016). Because such penalties directly constrain parameter values, they are expected to influence how task solutions are implemented in hidden-layer representations. However, neither penalty contributed meaningfully to the principal components, indicating that L1 and/or L2 regularization did not structure the dominant axes of representational variation.

One possible concern is that this might reflect selective survival under the 90% accuracy criterion—for example, if regularized networks were less likely to converge and therefore underrepresented in the analyzed sample. The present data provide no support for that interpretation. Among the 1504 successful networks, regularization was nearly perfectly balanced: 50.6% included L1 and 49.4% did not, while 49.7% included L2 and 50.3% did not. Because the hyperparameter grid was fully crossed and balanced, these counts suggest that success rates were comparable across regularization conditions (≈ 33 – 34%). This lack of performance differences aligns with prior findings that explicit regularization is not always necessary for successful generalization in overparameterized neural networks (Zhang et al., 2017).

Given that this explanation is unlikely, the absence of any detectable influence on representational geometry

is notable. Because L1 and L2 penalties act directly on hidden-layer weights, they are assumed to shape internal representations. Instead, the present findings suggest that, under these conditions, representational organization is largely determined by other factors, such as architectural constraints or optimization dynamics, rather than by explicit norm penalties alone. Several interpretations are possible. The regularization strengths tested may have been too small to substantially restrict the space of viable representational solutions, or the task itself may impose structural constraints that dominate any bias introduced by weight penalties. These interpretations remain speculative and require empirical verification. Distinguishing between these possibilities would require systematic manipulation of regularization strength together with direct analysis of both weight statistics and activation geometry.

Future Directions

Future work should examine whether the hyperparameter dependent representational variability observed here generalizes across tasks, architectures and learning regimes. The present findings were obtained in feedforward networks trained on a dual-task classification problem, and it remains unclear whether similar variability arises in architectures with different inductive biases, such as convolutional or transformer networks. Establishing such generality would clarify whether these effects reflect task-specific contingencies or more general properties of optimization. An important complementary direction concerns learning dynamics. Because the present analyses focused on converged representations, it remains unknown how hyperparameters shape representational trajectories during training. Tracking representational evolution could reveal whether different final geometries emerge from distinct developmental pathways or from convergence to different stable solutions. Finally, clarifying the mechanisms linking hyperparameters to representational geometry remains a key challenge. Hyperparameters modulate optimization noise, capacity and gradient structure, yet how these factors translate into geometric transformations of representation space is not well understood. Addressing these questions will be critical for determining when artificial neural networks can serve as reliable models of biological representations.

Conclusion

This study demonstrates that neural networks trained on the same task can develop substantially different internal representations despite comparable performance. Such representational variability poses a challenge for RSA-based model-brain comparisons, which rely directly on representational geometry rather than output behavior. These findings highlight the limitations of relying on

performance metrics alone when selecting or interpreting models for representational analyses. Both training stochasticity and modeling choices contribute to this variability, limiting the interpretability and replicability of representational results. Collectively, the present results motivate a more cautious and transparent use of ANNs as representational models in cognitive neuroscience.

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